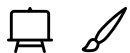


Applied Statistics: Time Series

Introduction to time series

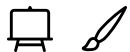
Robin Delabays

September 11, 2026



Definition of time series

According to you, what is a time series?

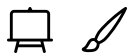


Definition of time series

According to you, what is a time series?

Two main ingredients:

1. **Series:** A “thing”, usually measurable, i.e., a number.
2. **Time:** This number changes over time.



Definition of time series

Formally, a time series is *a sequence of values each indexed by a time step*:

$$X = (x_{t_1}, x_{t_2}, \dots, x_{t_N}),$$

- x_t : the value at time t
- $t_1, t_2, \dots$ are the **time steps**.

Each element x_{t_i} of a time series is called a **sample**.

Time steps

For this course, we consider evenly spaced times steps.

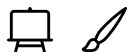
Therefore, it is enough to enumerate the components:

$$X = (x_1, x_2, \dots, x_N).$$

But we have to specify:

- the time of first recording t_0
- the **sampling rate**, i.e., the time between two consecutive samples.

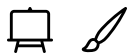
TODO: Add a figure here.



Examples of time series

- Any meteorological quantity, measured over time (temperature, pressure, wind speed,...)
- Stock market prices
- Any quantity measured inside a bioreactor (temperature, pH,...)
- Any health constant measured on a patient (temperature, heart rate, blood pressure,...)

TODO: Figures?



Acquisition - Time step

“Read” and “write” implies a gap between two consecutive values.

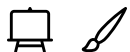
The time step should be:

- as large as possible...
- as small as necessary.

Example: *Recording sounds requires 20 000 samples per second, i.e., 20kHz.*

Notation: δ_t

Unit: second



Acquisition - Recording time

We cannot record for ever...

The recording time should be:

- as short as possible...
- as long as necessary.

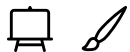
Example: *Recording a sound of less than 0.05s, sounds at 20Hz cannot be heard.*

Notation: T

Unit: second

Resampling

TODO: Do it on figures.



Representing

Recommendation: pyplot from matplotlib.

TODO: Make some code.

Noise

Example: *Measure the height of someone in millimeters.*

Noise

Example: *Measure the height of someone in millimeters.*

Data acquisition is imperfect.

Deviations from the “real” value are the **noise**.

Goal: Mitigate the effect of noise.

Hands on

TODO

